

BIH-VI Open Work Addendum

Residual uncertainties, completed stress tests, and next verification gates

Status: EFT-level completion stable; microscopic substrate not yet derived.

Executive Update

This addendum converts the remaining BIH-VI caveats into engineering-style scope boundaries and test gates. The Sqrt-Reg entropy background and EFT-clock realization remain viable. The project is not yet a UV-complete microscopic theory; it is a controlled, falsifiable EFT branch with a Planck-compatible observable island and a converged Mukhanov-Sasaki switch stress test.

Open work item	Current status	Result / remaining gap
EFT vs microscopic completion	Partly closed	A canonical scalar/EFT-clock realization exists. A UV-complete origin for the Sqrt-Reg entropy functional remains open.
Switch sensitivity	Partly closed	Delta=0.01 is pivot-safe and numerically converged in the M-S test. The switch profile is still phenomenological.
Single-clock limitation	Partly closed	The minimal $c_s=1$ branch has negligible pivot non-Gaussianity and no isocurvature by construction. Multi-field extensions remain untested.
Numerical splash resolution	Partly closed	The M-S feature is stable under step-size and initial-depth tests. Full Boltzmann transfer is not included.
Backreaction / loops	Preliminary pass	Feature amplitudes are small in absolute power; no PBH-scale enhancement appears in the first M-S stress test.
Curvature / anisotropy	Preliminary pass	Inflationary expansion suppresses shear strongly. Full Bianchi/curved-FLRW dynamics remain future work.

1. Microscopic Completion Gap

The original limitation remains: BIH-VI is not a first-principles UV-complete theory. However, the gap is smaller than before because the stable EFT-clock branch admits a canonical scalar realization. This shows that the Sqrt-Reg background can be carried by an ordinary stable scalar degree of freedom, even though it does not yet derive the entropy law from quantum gravity or holography.

$$H^2 = H_c^2 y(2-y), \quad z = 1-y, \quad m(y) = \mu z^3$$

$$\dot{\phi}^2 = 2 M_{\text{Pl}}^2 H_c^2 m(y) y(1-y)$$

$$V(y) = M_{\text{Pl}}^2 H_c^2 [3 y(2-y) - m(y)y(1-y)]$$

$$d\phi/dy = \pm M_{\text{Pl}} \sqrt{2(1-y) / [m(y)y^2(2-y)]}$$

For the cubic gear $m = \mu z^3$, the near-cap branch approaches a plateau. The potential is attractor-like: the Sqrt-Reg cap appears as a large-field plateau in canonical variables, while the exit occurs at finite field distance. This is structurally similar to the plateau logic of α -attractor models, but here the plateau is selected by the Sqrt-Reg entropy background rather than postulated as a supergravity potential.

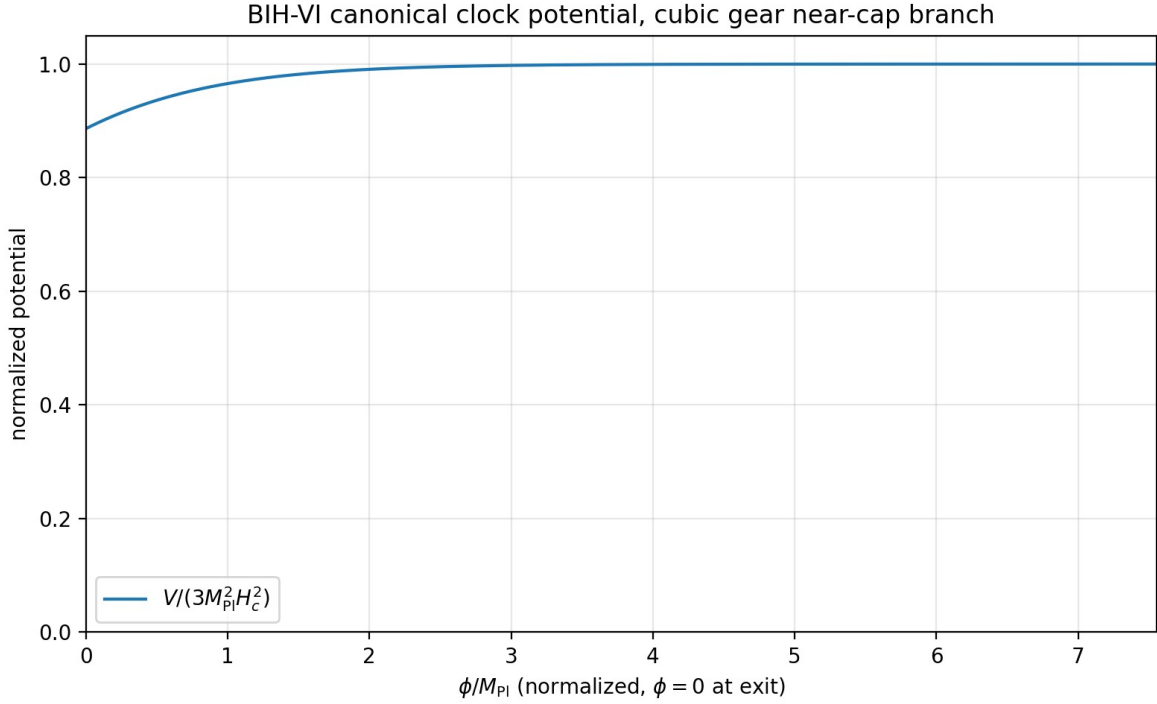


Figure 4. Canonical scalar realization of the cubic-gear BIH-VI clock potential. The plateau is the EFT representation of the Sqrt-Reg cap.

2. Non-Gaussianity Gate

At the CMB pivot, the benchmark branch uses $c_s = 1$ and $M_2^4 = 0$. Therefore the leading single-clock non-Gaussianity is slow-roll suppressed. The squeezed-limit consistency estimate gives $f_{\text{NL}}^{\text{local}} \sim 5(1-n_s)/12$, which is approximately $1.45e-2$ for $n_s = 0.9652$. This is far below current Planck constraints.

$$f_{\text{NL}}^{\text{local}} \sim 5(1-n_s)/12 \sim 1.5e-2$$

The switch can create feature-type non-Gaussianity near the exit scale. This is not yet computed; it requires cubic EFT terms and/or an in-in calculation through the switch. The M-S stress test shows no large scalar power enhancement, so there is no immediate PBH-like warning sign, but a feature-bispectrum calculation remains future work.

3. Backreaction and Loop Control

The pivot scalar amplitude remains $A_s \sim 2.1e-9$. The switch feature is localized near the exit. A slow-roll baseline estimate at the switch center gives $P_{\text{zeta}} \sim 1.8e-11$ before the M-S feature ratio. The converged M-S test found roughly a ~ 1.1 maximum ratio near $K \sim 1$ and suppression to ~ 0.05 in the exit-affected region. Thus the absolute switch-scale power is estimated to stay in the range:

$$P_{\text{zeta}}^{\text{switch}} \sim 9e-13 \text{ to } 2e-11 \quad (\text{benchmark estimate})$$

This is many orders of magnitude below the level usually associated with primordial-black-hole production or large perturbative backreaction. Loop corrections scale with powers of P_{zeta} and are therefore negligible at the pivot and likely negligible across the switch in this first EFT branch. A full cubic calculation is still required before declaring loop safety at all scales.

4. Curvature and Anisotropy Robustness

BIH-VI is not a contracting bounce; it is an expanding Sqrt-Reg shield followed by an inflationary EFT-clock window. That distinction matters. The usual BKL-style anisotropy danger is most severe in contracting nonsingular-bounce settings, where shear can grow as a^{-6} . In the BIH-VI expanding branch, shear is diluted during the inflationary window.

$$\rho_{\text{shear}} \propto a^{-6}, \quad \text{suppression over } N \text{ e-folds: } \exp(-6N)$$

$$N=55 \rightarrow \exp(-330) \sim 10^{-143}$$

Therefore any subdominant initial shear is exponentially erased during the benchmark window. What remains open is the fully coupled Bianchi-I version of the Sqrt-Reg/EFT-clock background. A future anisotropic calculation should check whether the Sqrt-Reg cap itself is stable to finite initial shear before the inflationary dilution phase.

5. Switch Sensitivity and M-S Convergence

The switch is phenomenological. It is a controlled valve rather than a derived substrate transition. The first stress test found $\Delta=0.01$ to be pivot-safe and numerically stable. The switch remains the most important phenomenological sensitivity because it determines the localized exit-scale feature.

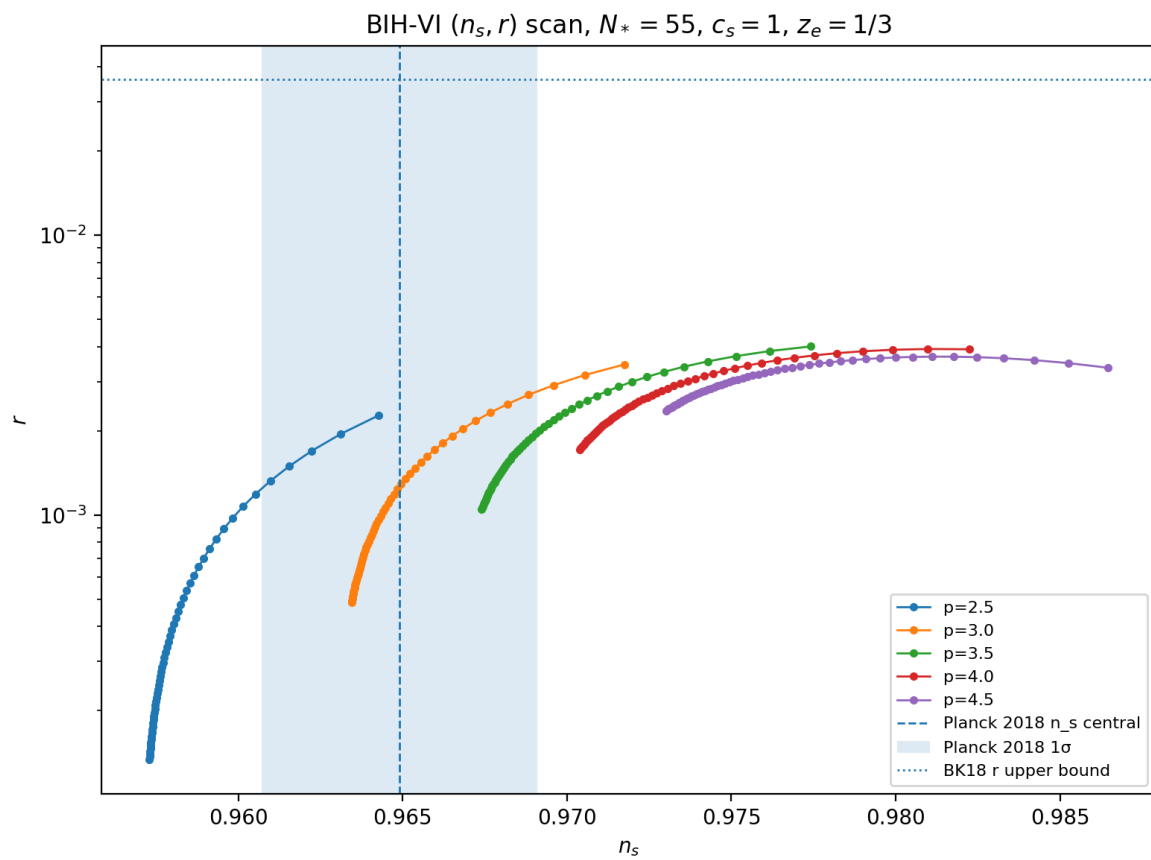


Figure 1. BIH-VI parameter island in the (n_s, r) plane.

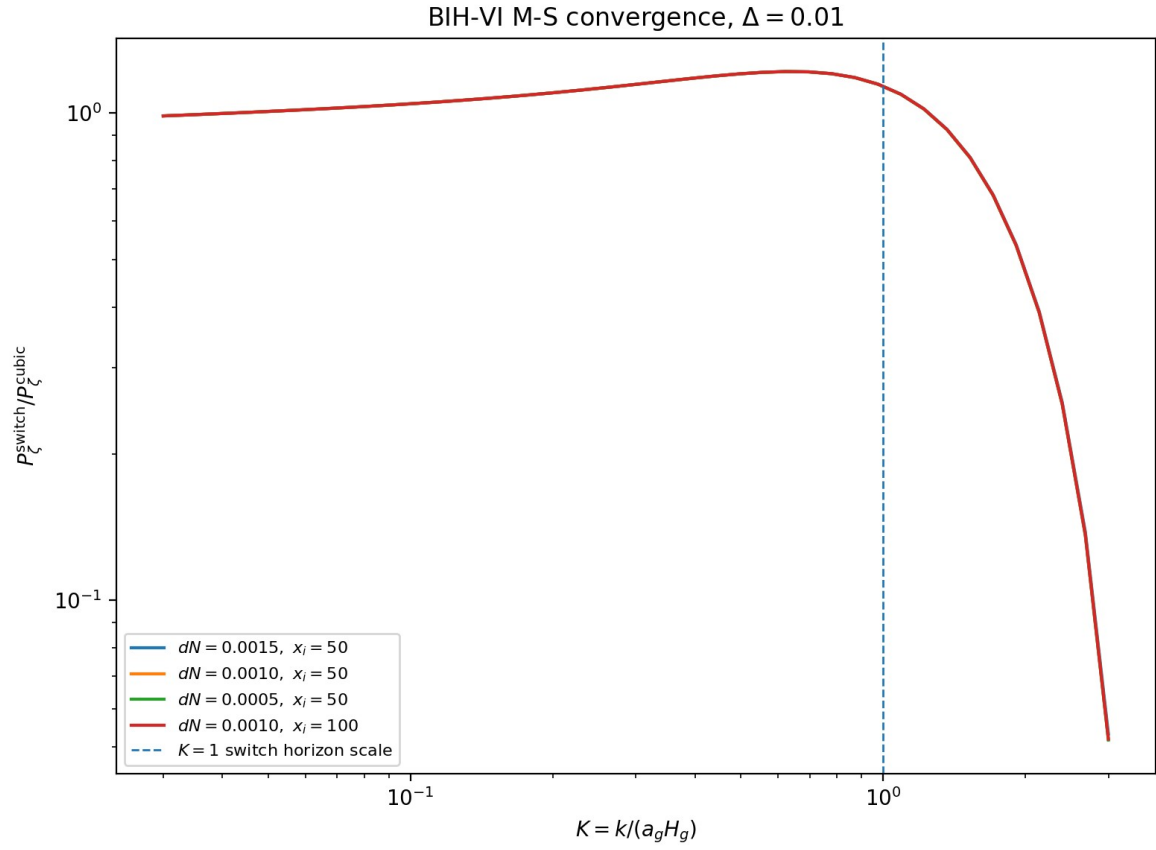


Figure 2. Mukhanov-Sasaki convergence test for the switch feature, Delta=0.01.

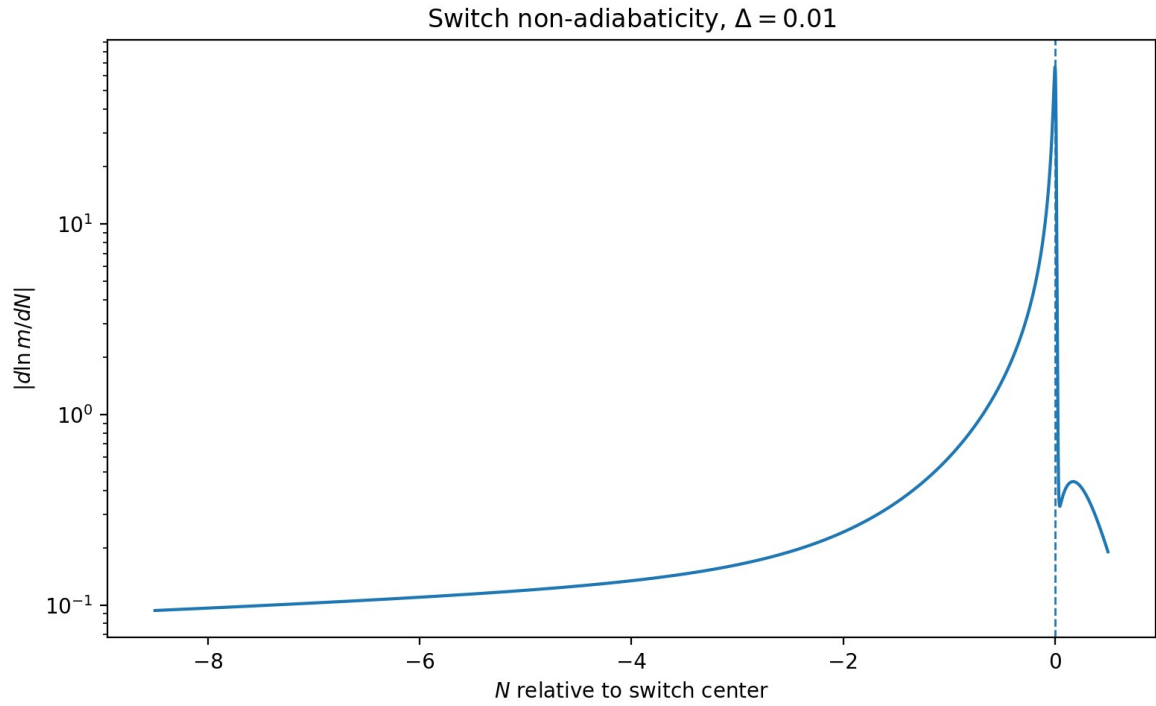


Figure 3. Switch non-adiabaticity, showing localized exit turbulence.

6. Boltzmann Transfer Scope Boundary

The M-S calculation gives the primordial curvature power proxy $P_{\zeta}(k)$. It does not directly generate CMB temperature and polarization spectra. A full CLASS/CAMB-style Boltzmann transfer calculation is required to convert the primordial feature into C_l predictions. This remains outside the present addendum.

7. Scope-Boundary Statement for the Main Report

Recommended wording:

This work establishes an EFT-level stability and phenomenology envelope for the BIH-VI Sqrt-Reg branch. It does not claim a unique microscopic UV completion of the entropy functional, does not treat multi-field/isocurvature dynamics, and does not compute full Boltzmann transfer functions. The contribution is the demonstration that a stable single-clock branch exists, overlaps current CMB constraints, and possesses a controlled exit-scale feature under Mukhanov-Sasaki evolution.

8. Closure Matrix

Question	Current answer
Can BIH-VI be stable at quadratic order?	Yes, in EFT-clock branch with $c_s^2 > 0$ and $\epsilon_H > 0$.
Does the pivot region match CMB constraints?	Yes, parameter island centered near $p=3$, $\mu \sim 1$.
Does the exit switch contaminate pivot modes?	No for $\Delta \sim 0.01$.
Does the switch feature converge numerically?	Yes under dN and initial-depth tests performed.
Is UV completion solved?	No. Canonical/EFT realization exists, but microscopic entropy origin remains open.
Is non-Gaussianity solved?	Pivot estimate is safely small; switch bispectrum remains open.
Is full CMB transfer solved?	No. M-S power is a direct primordial proxy only.
Is anisotropy solved?	Inflation dilutes shear strongly; full anisotropic background remains open.

References / Anchors

- Cheung et al., The Effective Field Theory of Inflation, arXiv:0709.0293.
- Maldacena, Non-Gaussian features of primordial fluctuations in single-field inflationary models, arXiv:astro-ph/0210603.
- Planck Collaboration, Planck 2018 Results IX: Constraints on primordial non-Gaussianity, arXiv:1905.05697.
- Kallosh and Linde, Superconformal inflationary alpha-attractors and related alpha-attractor literature.

- BKL/anisotropy caution: anisotropy growth in nonsingular bounce scenarios is a known limitation of contracting bounce models.

Prepared as an addendum to BIH-VI Technical Report.